

LOW ACRYLONITRILE NBR PATENT LANDSCAPE REPORT

1. ABSTRACT

This report reviews the provided patent set for evidence relevant to low acrylonitrile nitrile butadiene rubber (NBR). The target is a compositionally modified NBR grade with reduced acrylonitrile content, typically achieved by controlling butadiene/acrylonitrile copolymerization feed ratio, polymerization conditions, and downstream formulation choices. In the supplied manifest, however, the two primary patents do not disclose direct low-ACN NBR synthesis routes. Instead, they show adjacent uses of NBR-derived materials in elastomeric downhole seals and in fiber-cord adhesion systems, both of which rely on NBR chemistry but are not focused on monomer incorporation control.

US10443339B2 discloses elastomeric components for bore tools and packers reinforced with carbon-based nanoplatelets, with explicit emphasis on high-temperature/high-pressure service and chemical resistance. The patent mentions NBR among suitable elastomers in the broader background, but its inventive contribution is reinforcement of elastomeric matrices rather than NBR polymerization or acrylonitrile-content tuning. US11131058B2 discloses a cord manufacturing method using a first treatment agent containing a resorcin-formaldehyde condensate, a carboxy-modified latex, and a polycarbodiimide curing agent; the latex may be a carboxy-modified acrylonitrile-butadiene copolymer latex. This patent is relevant as a formulation/process disclosure involving NBR latex in adhesion treatment, but again not as a synthesis patent for low-ACN NBR.

Overall, the landscape in the provided evidence is formulation- and application-oriented rather than monomer-feed-oriented. The major process trends are: use of NBR as a component in reinforced elastomeric systems; use of carboxy-modified NBR latex in aqueous adhesive treatments; and emphasis on durability, flexibility, and environmental resistance. No patent in the manifest explicitly discloses a low-acrylonitrile NBR copolymerization recipe, acrylonitrile wt% target, or monomer feed ratio for producing low-ACN NBR.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US10443339B2
- Patent Title: Elastomeric materials

- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent does not disclose NBR polymerization or low-acrylonitrile synthesis. It describes forming an elastomeric component by mixing an elastomeric matrix with carbon-based nanoplatelets, then fitting the component onto bore-tool or packer hardware for downhole service. - Carbon-based nanoplatelets loading: less than approximately 10 percent by weight — Claim 1 - Temperature resistance context: about 160 degrees C. for HNBR — Detailed description - Temperature resistance context: about 220 degrees C. for AFLAS™ fluororubber — Detailed description - HPHT environment pressure: up to about 138 MPa — Detailed description - HPHT environment temperature: up to about 205 degrees C. — Detailed description - Ultra-HPHT environment pressure: up to about 241 MPa — Detailed description - Ultra-HPHT environment temperature: up to about 260 degrees C. — Detailed description

Relevant Experimental Evidence - Example 1: No structured example data was provided; the patent is primarily a disclosure of elastomeric downhole components. - Example 2: No structured example data was provided; the patent emphasizes carbon-based nanoplatelet-reinforced elastomeric materials. - Example 3: No structured example data was provided; the patent discusses packer assemblies and bore tools rather than NBR synthesis.

Technical Relevance This patent is only indirectly relevant because it identifies NBR as one of several elastomer families used in harsh-service downhole applications. It does not advance low-acrylonitrile NBR synthesis, but it shows the performance context in which NBR-based elastomers may be deployed.

Patent 2

- Patent Number: US11131058B2
- Patent Title: Method for manufacturing core wire for transmission belt, treatment agent, and ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a multi-step treatment process for power-transmission-belt cords using aqueous resorcin-formaldehyde-latex

type treatment agents. The key NBR-related disclosure is that the rubber component can include a carboxy-modified acrylonitrile-butadiene copolymer latex, used with a polycarbodiimide curing agent and water as the hydrophilic solvent. - Condensate (A1): 10 to 100 parts by mass relative to 100 parts by mass of rubber component (A2) — Claim 1 - Polycarbodiimide resin NCN equivalent: 300 to 450 per mole of carbodiimide group — Claim 1 - Curing agent (A3): 1 to 10 parts by mass relative to 100 parts by mass of rubber component (A2) — Claim 2 / synthesis section - Rubber composition (A): 15% to 20% by mass of the first treatment agent — synthesis section - Condensate (A1): 1 to 100 parts by mass relative to 100 parts by mass of rubber component (A2) — synthesis section - First treatment agent immersion time: 10 seconds — Example 4 / adhesion treatment description - Second treatment agent stirring time: 10 minutes — Preparation of second treatment agent - Second treatment agent total solid content concentration: 5% by mass — Preparation of second treatment agent - Untreated twisted yarn cord twist number: 20 to 300 times/m — synthesis section - Twist factor (T.F.): 0.01 to 10 — synthesis section - Twist factor (T.F.) preferred range: 1 to 6 — synthesis section - Twist factor (T.F.) preferred range: 0.2 to 2 — synthesis section

Relevant Experimental Evidence - Example 1: Demonstrated a first treatment agent containing COOH-modified NBR, RF condensate, polycarbodiimide dispersion, and water. - Example 2: Demonstrated an alternative blend using COOH-modified NBR with blocked isocyanate dispersion and water. - Example 3: Demonstrated use of unmodified NBR in an RFL-type blend with polycarbodiimide dispersion. - Example 4: Demonstrated a COOH-modified NBR-based treatment agent with blocked isocyanate dispersion and 5% total solid content in a related formulation table. - Example 9: Demonstrated a formulation using COOH-modified VP or COOH-modified NBR variants with different curing/additive packages.

Technical Relevance This patent is relevant because it explicitly uses carboxy-modified acrylonitrile-butadiene copolymer latex in an aqueous treatment system, showing an industrial NBR latex formulation context. It does not disclose low-ACN NBR synthesis, but it provides process and formulation parameters for NBR-derived latex use in adhesion applications.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Both patents are application/formulation patents rather than direct low-ACN NBR synthesis patents.
- US10443339B2 focuses on reinforced elastomeric downhole components using carbon-based nanoplatelets; US11131058B2 focuses on aqueous cord-treatment formulations using carboxy-modified NBR latex.
- US10443339B2 discloses a nanoplatelet loading below approximately 10 wt% and service temperatures up to 205-260 degrees C. in HPHT contexts; US11131058B2 discloses treatment-agent composition ranges such as 15-20 wt% rubber composition and 1-10 parts by mass curing agent per 100 parts rubber component.

- US11131058B2 is the only patent in the set that explicitly names acrylonitrile-butadiene copolymer latex, but it does not specify acrylonitrile content or monomer feed ratio.
- Neither patent provides a copolymerization recipe, initiator system, emulsifier system, chain-transfer agent, or acrylonitrile wt% target for producing low-acrylonitrile NBR.

4. CONCLUSION

The supplied patent set does not contain a direct disclosure of low-acrylonitrile NBR synthesis. The closest evidence is US11131058B2, which uses carboxy-modified acrylonitrile-butadiene copolymer latex in an aqueous RFL-type treatment system, and US10443339B2, which references NBR as a service elastomer in reinforced downhole components. For low-ACN NBR synthesis, the evidence base here is insufficient to recommend a specific polymerization recipe; however, the broader patent context supports aqueous emulsion-based NBR processing and formulation control. Based on the target definition, the most relevant synthesis parameters to seek in additional patents would be butadiene/acrylonitrile feed ratio, acrylonitrile content below about 20 wt% (often 10-18 wt%), emulsion polymerization conditions, and molecular-weight control via chain-transfer management.

5. REFERENCES

- US10443339B2 | Elastomeric materials | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10443339B2/en>
- US11131058B2 | Method for manufacturing core wire for transmission belt, treatment agent, and ... | Not disclosed | US | 2021 | <https://patents.google.com/patent/US11131058B2/en>